STDD: Building a Multimodal Spatiotemporal Expert for Zero-shot Action Recognition with CLIP

Anonymous Authors

Institution Withheld for Review

Abstract

Zero-shot action recognition requires recognizing unseen actions from text while preserving both static appearance cues and temporal dynamics. Image-text pretraining gives CLIP strong spatial semantics, but directly transferring it to videos leaves two gaps: limited temporal reasoning in the visual encoder and underspecified action text that often omits objects, motion order, and sub-actions. We present STDD, a multimodal spatiotemporal adaptation of CLIP for zero-shot action recognition. On the vision side, STDDinserts a parameter-free Space-time Cross Attention module into CLIP's ViT blocks through window shift masking, multi-scale channel mixing, spatial padding, and residual fusion. On the language side, STDDbuilds an Action Semantic Knowledge Graph and derives spatial and temporal prompts with a two-stage LLM pipeline. The model classifies videos with symmetric frame-prompt alignment and can optionally regularize adapted features with frozen-CLIP distillation. Using CLIP initialization and Kinetics-400 training, STDDreaches 93.4±2.2 and 88.6 on UCF101 and 68.7±4.5 and 58.7 on HMDB51 under EP1 and EP2 with ViT-L/14. Under EP3 with ViT-B/16, it achieves 85.2±1.2 on UCF101, 55.9±0.2 on HMDB51, and 75.1±0.7 on Kinetics-600, outperforming recent CLIP-based baselines.

1 Introduction

Zero-shot action recognition (ZSAR) aims to recognize actions whose labels are unavailable during supervised training. The problem is particularly appealing for video understanding because real-world action vocabularies evolve quickly, while exhaustively collecting annotated videos for every new activity is expensive. Recent vision-language models, especially CLIP (Radford et al. 2021), make ZSAR practical by embedding visual observations and language into a shared space. However, CLIP is trained on still image-text pairs, so its strengths are mostly spatial: it can identify objects, scenes, and coarse human poses, but it is less reliable when action categories differ primarily in motion or sub-action order.

Figure 1 illustrates the two failure modes we target. First, static appearance alone is ambiguous: similar objects or backgrounds can support different actions. Second, temporal evidence alone is also ambiguous when motion phases or action order are subtle. Existing CLIP-based video transfer methods usually address only one side of this problem. Some focus on visual adaptation by adding temporal modules (Ni et al. 2022; Weng et al. 2023; Wu et al. 2024; Huang et al. 2024). Others focus on richer text descriptions or prompt learning (Wu et al. 2023; Chen et al. 2024; Kahatapitiya et al. 2024; Jia et al. 2024). In our view, strong ZSAR needs both: cross-frame visual interaction and structured action semantics that explicitly separate what appears from how it unfolds over time.

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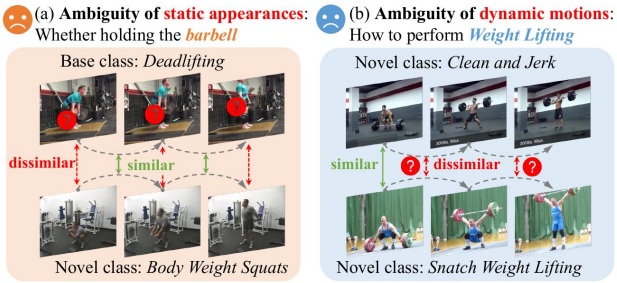

Figure 1: Without collaborative multimodal spatiotemporal understanding, zero-shot action recognition is hindered by ambiguity from static appearances and dynamic motions. Similar objects can correspond to different actions, and visually similar frames can still represent distinct motion sequences.

We therefore propose STDD (**S**patiotemporal **D**ynamic **D**uo), a CLIP-based framework that jointly upgrades the visual and textual branches. In the visual encoder, we replace each CLIP residual block with a parameter-free Space-time Cross Attention (STCA) block that uses window shift masking to expose complementary spatial tokens across frames, multi-scale channel mixing to exchange temporal cues, and spatial padding to restore the original layout. In the text encoder, we construct an Action Semantic Knowledge Graph (ASKG) for each action and derive two families of prompts: spatial prompts describing objects and static relations, and temporal prompts describing ordered sub-actions. We then classify videos with symmetric frame-prompt alignment, which matches frames to the most relevant prompts and prompts to the most relevant frames.

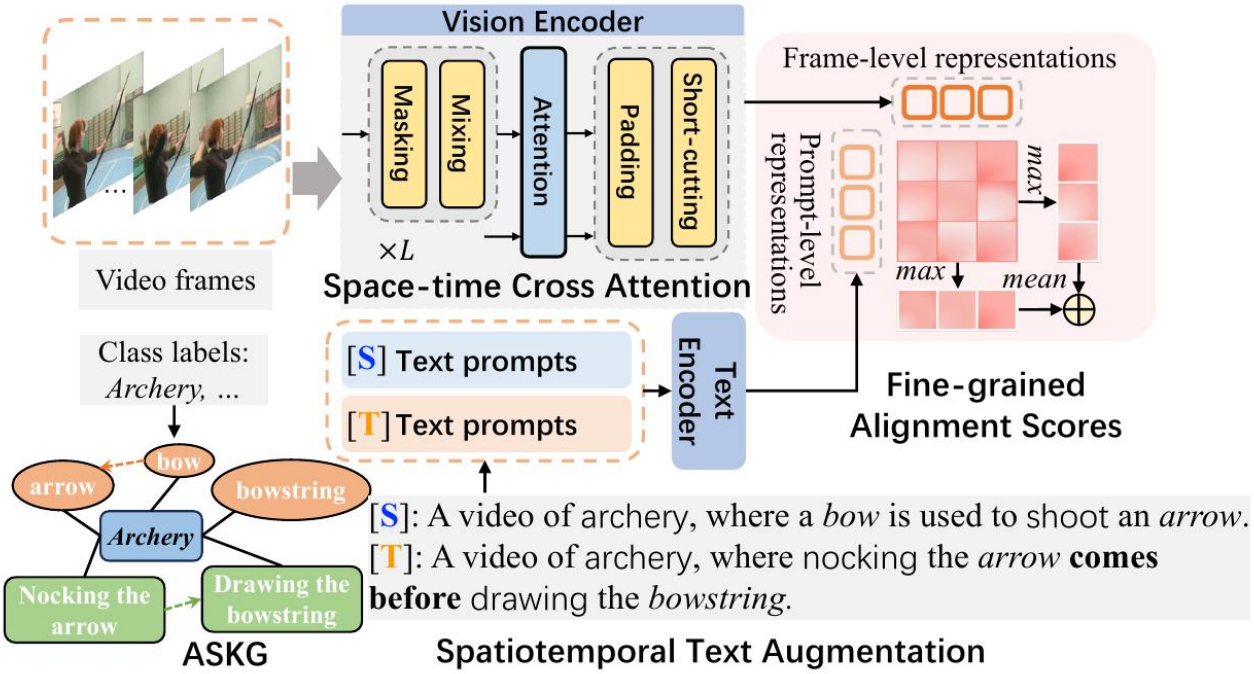

Figure 2: Overview of STDD. We adapt CLIP in two complementary ways. The visual branch performs parameter-free space-time interaction inside the ViT through masking, channel mixing, attention, padding, and residual fusion. The text branch constructs an ASKG and derives spatial prompts $[S]$ and temporal prompts $[T]$. Video classification uses symmetric frame-prompt alignment over the resulting fine-grained similarity matrix.

The contributions of this work are:

- We introduce STCA, a parameter-free visual adaptation that turns CLIP's image-centric transformer into a spatiotemporal encoder without introducing extra temporal parameters.
- We propose ASKG-driven spatiotemporal text augmentation that decomposes action semantics into spatial cues and temporal sub-actions, producing richer zero-shot prompts than category names alone.
- We combine both branches through symmetric frame-prompt alignment and show state-of-the-art ZSAR performance on UCF101, HMDB51, and Kinetics-600 across standard evaluation protocols.

2 Related Work

Zero-shot Action Recognition Before Vision-Language Models

Earlier ZSAR methods mostly relied on manually designed semantics, word embeddings, or knowledge transfer from seen classes. Representative approaches include knowledge-graph reasoning with TS-GCN (Gao, Zhang, and Xu 2019), generative synthesis in CWEGAN (Mandal et al. 2019), elaborative descriptions in ER-ZSAR (Chen and Huang 2021), and dynamic action signatures in DASZL (Kim et al. 2021). Cross-modal encoders for joint video-text learning also improved classical ZSAR (Lin et al. 2022). These methods established the importance of semantic transfer, but they predate large-scale image-text pretraining and therefore lack the broad visual-language prior available in CLIP (Radford et al. 2021).

Adapting CLIP to Video

With CLIP as initialization, recent work has explored how to preserve generalization while injecting temporal reasoning. ActionCLIP (Wang, Xing, and Liu 2021) frames video recognition as video-text matching with prompt engineering. X-CLIP (Ni et al. 2022) introduces cross-frame attention and video-specific prompts. Open-VCLIP and Open-VCLIP++ (Weng et al. 2023; Wu et al. 2024) treat CLIP-to-video transfer as a continual adaptation problem and emphasize robustness to unseen categories. BIKE (Wu et al. 2023), ViLT-CLIP (Wang et al. 2024), and FROSTER (Huang et al. 2024) further study cross-modal knowledge exploration, multi-modal prompting, and teacher regularization. Our method differs in two ways: it uses a parameter-free temporal mechanism inside each ViT block instead of adding new temporal parameters, and it couples this visual adaptation with structured spatiotemporal text augmentation.

Semantic Augmentation for Open-vocabulary Video Recognition

Several recent methods argue that richer text is essential for open-vocabulary or zero-shot video understanding. Text4Vis (Wu, Sun, and Ouyang 2023) transfers knowledge from text-derived classifier targets. OST (Chen et al. 2024) prompts

an LLM to produce spatiotemporal descriptors and matches them to frame features. VicTR (Kahatapitiya et al. 2024) conditions text on video evidence, while AP-CLIP (Jia et al. 2024) creates action-conditioned prompts from LLM knowledge. Our text branch is most closely related to these efforts, but instead of producing a flat set of descriptions, we explicitly organize action semantics into a graph with object entities, object relations, action-object relations, and ordered sub-actions. This lets us separate spatial prompts from temporal prompts and analyze their contributions independently.

3 Method

Problem Setup

We consider cross-dataset zero-shot action recognition. A model is trained on labeled videos from Kinetics-400 (Kay et al. 2017) and evaluated on unseen action classes from UCF101 (Soomro, Zamir, and Shah 2012), HMDB51 (Kuehne et al. 2011), or Kinetics-600 (Carreira et al. 2018). Let a video be sampled into T frames, and let the CLIP image encoder produce frame-level features $\{v_t\}_{t=1}^{T}$, where $v_t \in \mathbb{R}^d$. For each candidate class k, we build a set of prompt features $\{e_{k,p}\}_{p=1}^{P_k}$ from the text branch. The task is to score each unseen class using only these visual and textual representations.

Space-time Cross Attention

Figure 2 and Figure 3 summarize our visual adaptation. We start from CLIP's ViT and replace each standard residual attention block with STCA. Importantly, STCA reuses the original self-attention and MLP layers and adds only deterministic tensor operations for temporal interaction.

Consider token sequence $X \in \mathbb{R}^{(N+1)\times BT\times D}$ inside a ViT block, where N is the number of patch tokens per frame, B is the batch size, and the extra token is the class token. STCA performs four steps.

Window shift masking. Patch tokens are first reshaped into the spatiotemporal grid and partitioned with stride $(1, 2, 2)$, matching the released configuration. For each local spatial cell, STCA keeps complementary token subsets in neighboring frames by cyclically offsetting the mask pattern over time. With mask rate 0.5, each frame exposes only part of its local tokens, but adjacent frames reveal different positions. This yields a reduced token sequence $\widetilde{X}$ containing the class token and only the unmasked patch tokens.

Multi-scale channel mixing. Temporal information is injected by shifting different channel groups across time before attention. Specifically, the first channel groups are shifted forward and backward by temporal scales $\delta \in \mathcal{S}$, while the remaining channels stay unchanged. The code supports one or multiple scales; the provided ViT-B/16 setting uses $\mathcal{S} = \{1\}$, while the ViT-L/14 setting uses a larger temporal scale. For each δ,

$$Z_\delta = \mathrm{CM}_\delta(\widetilde{X}) + \mathrm{Attn}\big(\mathrm{LN}(\mathrm{CM}_\delta(\widetilde{X}))\big), \tag{1}$$

where CM_δ denotes channel mixing at temporal offset δ. The mixed outputs are averaged across scales:

$$\bar{Z} = \frac{1}{|\mathcal{S}|}\sum_{\delta\in\mathcal{S}} Z_\delta. \tag{2}$$

Spatial padding. In parallel, the original token sequence follows CLIP's standard spatial attention:

$$Y = X + \mathrm{Attn}(\mathrm{LN}(X)). \tag{3}$$

The mixed features $\bar{Z}$ are then written back to their original spatial locations through spatial padding, while masked positions retain the corresponding features from Y. This restores the full token layout after cross-frame interaction.

Residual fusion. Finally, the block applies the original MLP residual on top of the attention-updated tokens:

$$X' = \mathrm{Pad}(Y, \bar{Z}) + \mathrm{MLP}(\mathrm{LN}(Y)). \tag{4}$$

Because masking, mixing, and padding are parameter-free, STCA preserves CLIP's parameterization while enabling cross-frame information exchange.

ASKG-driven Spatiotemporal Text Augmentation

Text labels such as "archery" or "clean and jerk" are often too short to disambiguate fine-grained actions. We therefore build an Action Semantic Knowledge Graph (ASKG) for each action with a two-stage LLM pipeline implemented in the released code.

In the first stage, an LLM is prompted to extract: 1) object entities; 2) object-object relation triples; 3) action-object relation triples; 4) ordered sub-actions; and 5) action-action precedence relations. The resulting graph explicitly separates static concepts from temporal procedural knowledge. For example, the graph for *archery* contains objects such as bow, arrow, target, and quiver, together with ordered sub-actions such as gripping the bow, aiming, and releasing the arrow.

In the second stage, these triples are verbalized into natural-language prompts. We obtain three prompt types:

- object-object prompts (`xprompt_oo`), e.g., "where a bow is used to shoot an arrow";
- action-object prompts (`xprompt_ao`), e.g., "which requires a bow";
- action-action prompts (`xprompt_aa`), e.g., "where aiming at the target comes before releasing the arrow".

We treat object-object and action-object prompts as *spatial* prompts and action-action prompts as *temporal* prompts. Each completion is wrapped by hand-crafted templates such as "a video of a person {}." and encoded by the frozen CLIP text encoder. In practice, we use the union of spatial and temporal prompts (denoted *ST Aug.*) as the default text representation. This decomposition also matches the alignment rule in (6). Spatial prompts usually remain valid across many frames, whereas temporal prompts are only relevant to short motion phases. Keeping the two types explicit prevents temporal cues from being diluted by persistent object evidence and gives each max operator a cleaner set of candidates.

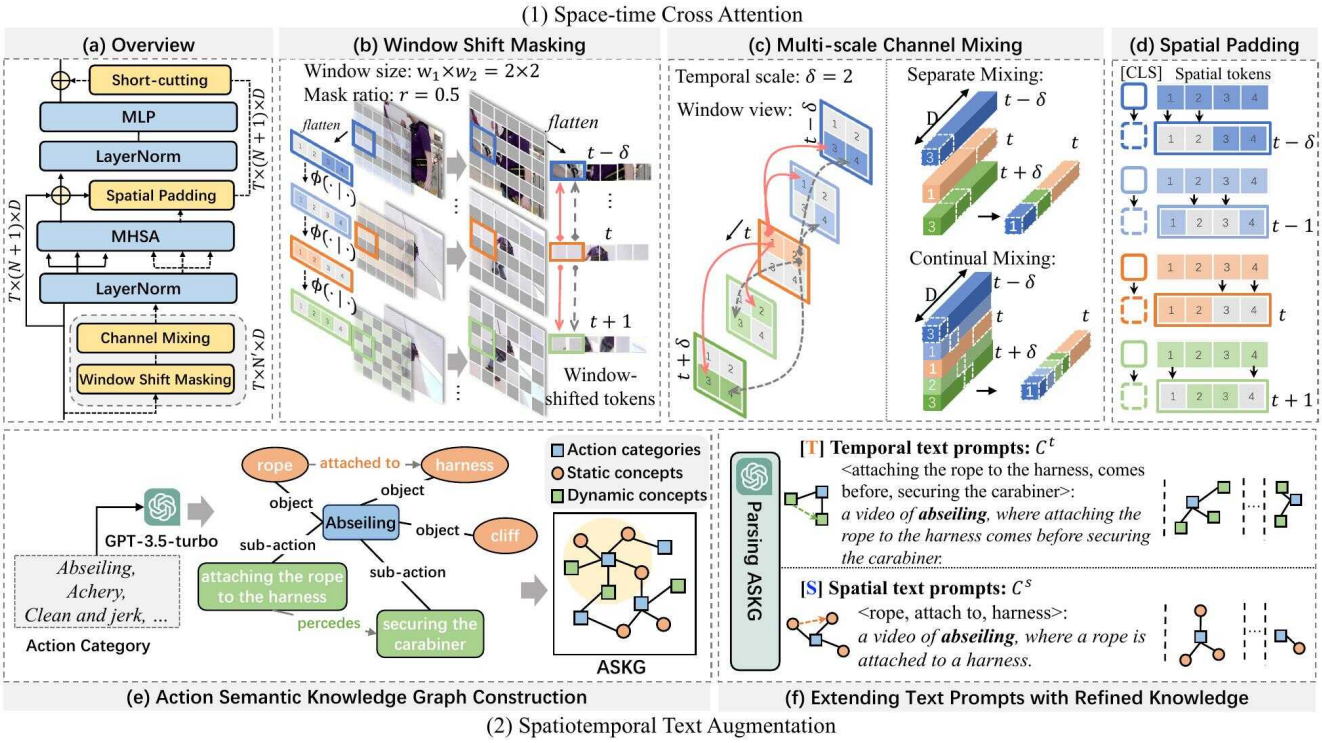

Figure 3: Detailed illustration of STDD. STCA performs four steps: window shift masking, multi-scale channel mixing, self-attention on the mixed tokens, and spatial padding back to the original token layout. In parallel, an LLM-derived ASKG is parsed into spatial prompts and temporal prompts that are encoded by CLIP's text encoder.

Symmetric Frame-prompt Alignment

Given frame features v_t and prompt features $e_{k,p}$ for class k, we compute a fine-grained similarity matrix

$$S^k_{t,p} = \tau\, v_t^\top e_{k,p}, \tag{5}$$

where τ is CLIP's learned logit scale. Instead of averaging frames or prompts uniformly, STDD uses the symmetric max-aggregation implemented in the code:

$$s_k = \frac{1}{2}\left(\frac{1}{T}\sum_{t=1}^{T}\max_p S^k_{t,p} + \frac{1}{P_k}\sum_{p=1}^{P_k}\max_t S^k_{t,p}\right). \tag{6}$$

The first term lets each frame choose its most relevant prompt; the second lets each prompt choose its most relevant frame. This symmetric design is important because some prompts describe localized visual states while others describe particular motion phases.

Training uses cross-entropy on the seen Kinetics-400 classes:

$$\mathcal{L}_{\text{cls}} = -\log \frac{\exp(s_y)}{\sum_k \exp(s_k)}. \tag{7}$$

The released training pipeline also supports an optional frozen-CLIP regularizer in the spirit of teacher preservation (Huang et al. 2024). Let $\hat{v}_t$ be the adapted feature and v_t^0 the frozen CLIP feature. The distillation term is

$$\mathcal{L}_{\text{dist}} = 1 - \frac{1}{T}\sum_{t=1}^{T}\cos(\hat{v}_t, v_t^0), \tag{8}$$

and the total loss is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{dist}}$ when this regularizer is enabled.

4 Experiments

Experimental Setup

We initialize the video encoder from CLIP ViT-B/16 or ViT-L/14 (Radford et al. 2021). Following the released configurations, training uses 8 frames per clip with 224×224 crops, AdamW, cosine learning-rate decay, and a default 12-epoch schedule for the STDD configurations. Test-time evaluation uses three temporal views and one spatial crop. For STCA, the default settings are mask rate 0.5, mask stride $(1, 2, 2)$, and channel fold 32. The provided prompt features use multiple hand-crafted video templates and ASKG-derived prompt sets. We report top-1 accuracy on the standard ZSAR protocols used by prior CLIP-based work (Weng et al. 2023; Wu et al. 2024; Chen et al. 2024; Huang et al. 2024). Because the available artifacts do not include hardware logs, we omit hardware details.

Comparison on UCF101 and HMDB51

Table 1 reports results on the EP1 and EP2 protocols for UCF101 and HMDB51. STDD with ViT-L/14 achieves the best accuracy in all four columns, reaching **93.4**±**2.2** and **88.6** on UCF101, and **68.7**±**4.5** and **58.7** on HMDB51. Compared with Open-VCLIP using the same ViT-L/14

backbone (Weng et al. 2023), the gains are modest but consistent: +0.3 on UCF EP1, +0.7 on UCF EP2, +0.2 on HMDB EP1, and +0.4 on HMDB EP2.

The ViT-B/16 model is also competitive. Relative to Open-VCLIP with ViT-B/16, STDD improves from 83.5 to 85.3 on UCF EP2 and from 53.2 to 54.7 on HMDB EP2. These gains are important because they are achieved with the smaller backbone and indicate that the proposed visual-textual coupling is not limited to the large model.

Comparison on EP3 and Kinetics-600

Table 2 compares against recent CLIP-based methods on the EP3 setting and Kinetics-600. Using ViT-B/16, STDD achieves the best result on all three benchmarks: **85.2±1.2** on UCF101, **55.9±0.2** on HMDB51, and **75.1±0.7** on Kinetics-600. The gains over FROSTER (Huang et al. 2024) are +0.4 on UCF and +0.3 on K600, while the advantage over Open-VCLIP++ (Wu et al. 2024) is larger on UCF and K600. This pattern suggests that combining temporal visual interaction with richer action semantics remains beneficial even when strong CLIP-based baselines already use LLM-generated text.

Effect of Spatiotemporal Text Augmentation

Table 3 isolates the text branch by comparing category text prompts (CT), OST's spatiotemporal descriptors (STD), and our ASKG-based spatiotemporal augmentation. The proposed ST Aug. achieves the best result on all three datasets, improving over CT by +1.4 on UCF101, +1.9 on HMDB51, and +1.6 on Kinetics-600. The margin over OST's descriptors is smaller, especially on K600, but still positive. This supports the claim that explicitly separating spatial and temporal knowledge is a better match for fine-grained video-text alignment than using a single undifferentiated prompt pool.

5 Analysis

Ablating STCA Operations

Table 4 studies the visual branch on UCF101. A frozen CLIP baseline reaches 74.2. Adding STCA with prompt-based adaptation already brings the model to 79.8–82.7, while full fine-tuning further improves the range to 84.5–85.2. Two trends are clear. First, window shift masking consistently helps: comparing rows with and without WSM at the same adaptation level shows gains of +2.2 to +2.5. Second, continual channel mixing is slightly better than separate mixing in every matched comparison. The best configuration combines full fine-tuning, WSM, and continual mixing for 85.2, an absolute +11.0 gain over the frozen baseline.

Why Fine-grained Alignment Helps

Figure 4 compares different text sources and matching rules on UCF101 EP2. Using only the class name ($[C]$) yields 83.81. Replacing it with spatial prompts and simple video-class matching increases performance to 85.25, showing that object-centric knowledge is already valuable. Temporal prompts also help, but less strongly on their own. The best result, 85.33, comes from combining spatial and temporal prompts with frame-prompt alignment. This is consistent with (6): prompts describing specific motion stages should not be forced to explain every frame, and frames containing static evidence should not be diluted by irrelevant temporal descriptions. The effect is also consistent with the dataset-level comparisons in Table 2. The largest gains appear on UCF101 and Kinetics-600, while HMDB51 improves by a smaller margin. We view this as evidence that structured prompts help most when the video contains recoverable object and phase cues; when clips are shorter or visually noisier, the same mechanism still helps, but the advantage is naturally compressed.

Qualitative Visualization

Figure 5 visualizes attention maps and prompt alignment scores for *Archery*. Compared with vanilla CLIP, STDD attends more consistently to the bow, arrow, and the actor's arm configuration, which are the key cues needed to distinguish archery from other visually similar actions. The frame-prompt alignment scores are also interpretable. A spatial prompt about using a bow to shoot an arrow obtains the highest score (0.73), while temporal prompts such as gripping the bow and aiming before release align strongly with the early and late frames respectively (0.68 and 0.57). This supports our central claim that ZSAR benefits from combining spatial evidence with temporally localized textual knowledge.

6 Conclusion

We presented STDD, a CLIP-based framework for zero-shot action recognition that upgrades both visual and textual reasoning. The visual branch uses parameter-free Space-time Cross Attention to exchange information across frames inside CLIP's ViT blocks. The text branch uses an Action Semantic Knowledge Graph to produce spatial and temporal prompts, and the model matches the two branches with symmetric frame-prompt alignment. Across UCF101, HMDB51, and Kinetics-600, the method consistently improves over strong CLIP-based baselines, while ablations and visualizations show that both the temporal visual adaptation and the structured text augmentation are necessary for the final gains.

Method	Venue	Encoder	SA Method[1]	UCF		HMDB	
				EP 1	EP 2	EP 1	EP 2
Uni-modal Vision Training							
TS-GCN (Gao, Zhang, and Xu 2019)	AAAI'19	GoogleNet	WE	36.14 ± 4.8	-	23.2	-
CWEGAN (Mandal et al. 2019)	CVPR'19	I3D	WE	26.9 ± 2.8	-	30.2	-
ER-ZSAR (Chen and Huang 2021)	ICCV'21	TSM	ED	51.8 ± 2.9	-	35.3 ± 4.6	-
DASZL (Kim et al. 2021)	AAAI'21	TSM	HA	48.9 ± 5.8	-	-	-
Adapting Pretrained VLMs							
BIKE (Wu et al. 2023)	CVPR'23	ViT-L/14	AT	86.6 ± 3.4	80.8	61.4 ± 3.6	52.8
Text4Vis* (Wu, Sun, and Ouyang 2023)	AAAI'23	ViT-L/14	ST Aug.	89.5 ± 2.9	84.2	63.7 ± 3.2	52.9
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-B/16	CT	89.9 ± 1.7	83.5	64.5 ± 4.5	53.2
Open-VCLIP (Weng et al. 2023)	ICML'23	ViT-L/14	CT	93.1 ± 1.9	87.9	68.5 ± 4.0	58.3
ViLT-CLIP (Wang et al. 2024)	AAAI'24	ViT-B/16	PE	-	73.9	-	45.3
STDD	-	ViT-B/16	ST Aug.	90.3 ± 1.7	85.3	64.7 ± 3.8	54.7
STDD	-	ViT-L/14	ST Aug.	**93.4 ± 2.2**	**88.6**	**68.7 ± 4.5**	**58.7**

[1] SA: semantic augmentation; WE: word embeddings; ED: elaborative descriptions; HA: hand-crafted attributes; CT: category text prompts; AT: attribute text prompts; PE: prompt embeddings.

Table 1: Comparison on UCF101 and HMDB51 under EP1 and EP2. "*" denotes our re-evaluation.

Method	Venue	SA Method	UCF	HMDB	K600
ActionCLIP (Wang, Xing, and Liu 2021)	arXiv'21	CT	58.3 ± 3.4	40.8 ± 5.4	66.7 ± 1.1
X-CLIP (Ni et al. 2022)	ECCV'22	CT	72.0 ± 2.3	44.6 ± 5.2	65.2 ± 0.4
MAXI (Lin et al. 2023)	ICCV'23	VC[1]	78.2 ± 0.7	52.3 ± 0.6	71.5 ± 0.8
VicTR (Kahatapitiya et al. 2024)	CVPR'24	CT + AT	72.4 ± 0.3	51.0 ± 1.3	-
OST (Chen et al. 2024)	CVPR'24	STD[1]	77.9 ± 1.3	54.9 ± 1.1	73.9 ± 0.8
AP-CLIP (Jia et al. 2024)	ACM MM'24	AP[1]	82.4 ± 0.5	55.4 ± 0.8	73.4 ± 1.0
Open-VCLIP++ (Wu et al. 2024)	TPAMI'24	VC	83.9 ± 0.6	55.6 ± 1.4	73.4 ± 0.8
FROSTER (Huang et al. 2024)	ICLR'24	AD[1]	84.8 ± 1.1	54.8 ± 1.3	74.8 ± 0.9
STDD	-	ST Aug.	**85.2 ± 1.2**	**55.9 ± 0.2**	**75.1 ± 0.7**

[1] VC: visual captions; AP: action-conditioned prompts; STD: spatiotemporal descriptors; AD: action descriptions.

Table 2: Comparison with recent CLIP-based methods on UCF101, HMDB51, and Kinetics-600 under EP3. All methods use CLIP ViT-B/16.

SE Method	**UCF**	**HMDB**	**K600**
CT (Wang, Xing, and Liu 2021)	83.8	54.0	73.5
STD (Chen et al. 2024)	84.9	55.1	75.0
STDD ST Aug.	**85.2**	**55.9**	**75.1**

Table 3: Effect of different semantic augmentation strategies on EP3.

Adaptation	**WSM**	**MCM**	**UCF**
Frozen	✗	✗	74.2
Prompt-based	✗	Separate	79.8 (+5.6)
Prompt-based	✗	Continual	80.1 (+5.9)
Prompt-based	✓	Separate	82.3 (+8.1)
Prompt-based	✓	Continual	82.7 (+8.5)
Fully-tuned	✗	Separate	84.5 (+10.3)
Fully-tuned	✗	Continual	84.6 (+10.4)
Fully-tuned	✓	Separate	85.0 (+10.8)
Fully-tuned	✓	Continual	**85.2 (+11.0)**

Table 4: Effect of STCA components on UCF101. WSM denotes window shift masking and MCM denotes multi-scale channel mixing.

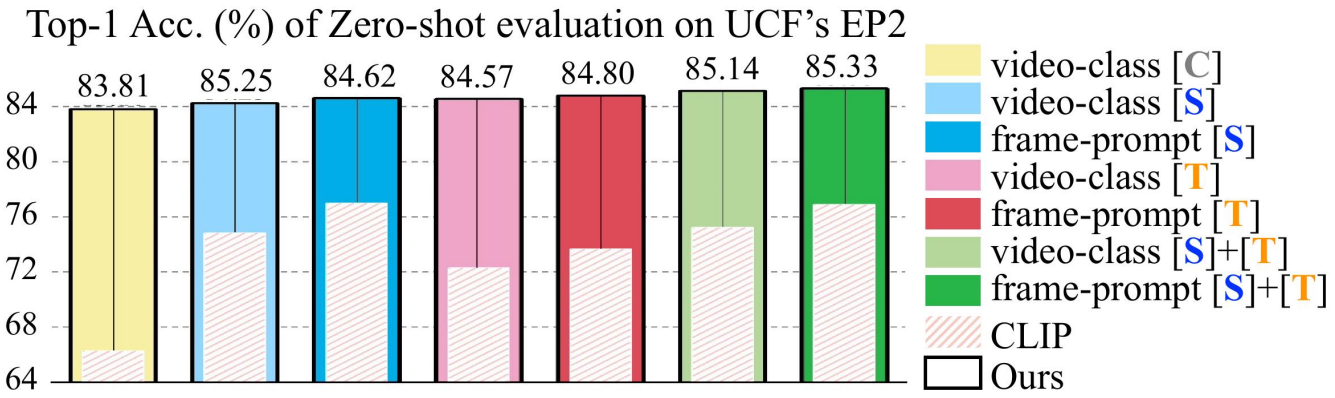

Figure 4: Effect of prompt type and matching granularity on UCF101 EP2. $[C]$ uses only class names, $[S]$ uses spatial prompts, and $[T]$ uses temporal prompts. Frame-prompt alignment with both prompt types performs best.

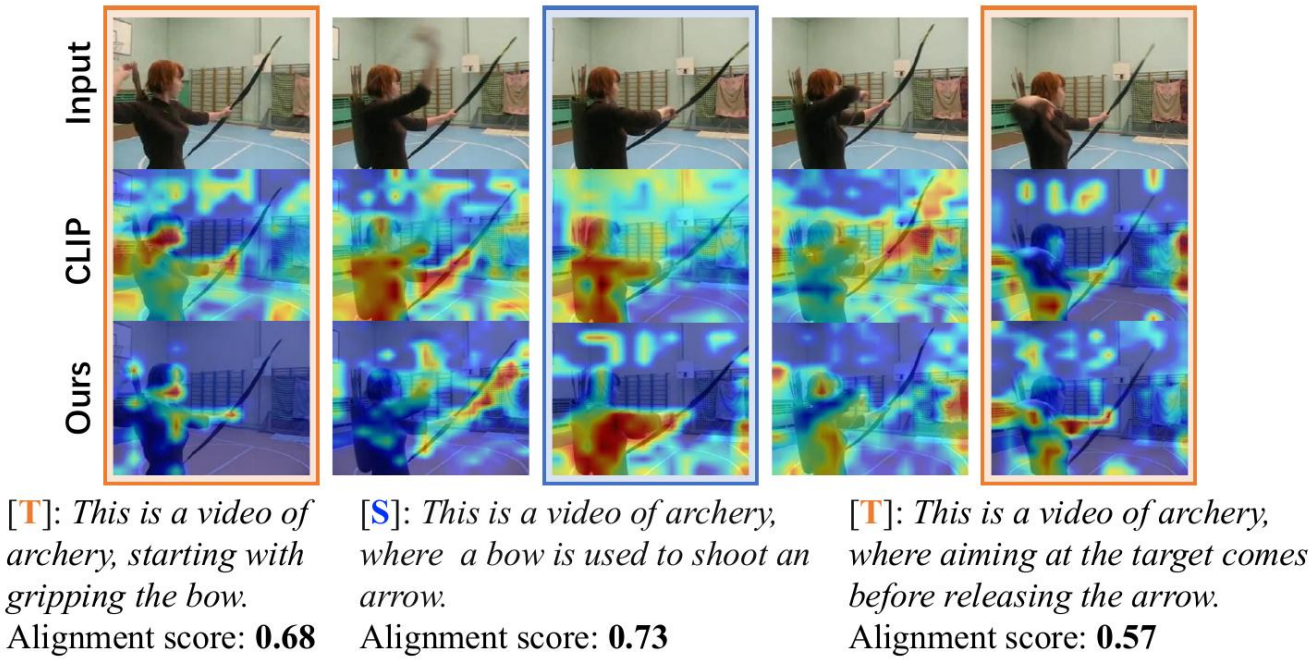

Figure 5: Qualitative comparison on *Archery*. Our model focuses more tightly on action-relevant regions than vanilla CLIP and produces interpretable prompt alignments for both spatial and temporal descriptions.

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